Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery - Personalized Sign Language Tutoring System for Caregivers

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Project Proposal Report

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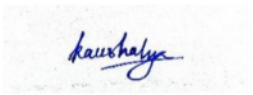
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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The increasing demand for effective caregiving, combined with the growing population of deaf and hard-of-hearing individuals, has highlighted critical communication barriers in healthcare settings. Caregivers often lack sufficient sign language skills, leading to misunderstandings, reduced care quality, and decreased patient satisfaction. Despite advancements in sign-language technologies, most existing solutions focus on general vocabulary learning or gesture recognition rather than practical, caregiver-specific skill development.

A key research gap exists in the absence of an integrated, personalized tutoring system that supports adaptive learning, interpretable feedback, and real-time performance evaluation tailored to caregiving contexts. Current systems provide limited personalization, unclear corrective guidance, and fail to account for variations in user learning pace, making them inadequate for real-world application.

To address this gap, this research proposes a Personalized Sign Language Tutoring System for Caregivers. The system integrates computer vision-based gesture capture, speed-tolerant evaluation using Dynamic Time Warping (DTW), interpretable feedback generation, and 3D avatar-based visual guidance within a unified web-based platform. Additionally, adaptive learning mechanisms personalize practice sessions based on user performance and recurring errors, enhancing skill acquisition and retention. The system will be developed using technologies such as Python, MediaPipe, OpenCV, React JS, Three.js, and Flask, supported by a lightweight SQLite database. These tools enable real-time interaction, accurate gesture analysis, and scalable deployment.

Expected outcomes include improved caregiver sign language proficiency, enhanced communication accuracy, and increased confidence in real-world caregiving scenarios. Socially, the system promotes inclusive healthcare and reduces communication inequalities, while commercially, it offers scalable opportunities through institutional deployment and Software-as-a-Service (SaaS) models.

Keywords: Sign Language Tutoring, Caregiver Training, Computer Vision, Adaptive Learning, Human-Computer Interaction

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ASL	American Sign Language
DSR	Design Science Research
DTW	Dynamic Time Warping
HCI	Human-Computer Interaction
UI	User Interface
UX	User Experience
SaaS	Software as a Service
SUS	System Usability Scale
SL	Sign Language
CV	Computer Vision
3D	Three-Dimensional
SDG	Sustainable Development Goals

Table 1 List of Abbreviations

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1. INTRODUCTION

1.1. Background and Context

Effective communication is a fundamental requirement in caregiving, as it directly influences patient safety, comfort, and the overall quality of care. Caregivers are responsible for assisting individuals with daily activities, monitoring health conditions, and responding to both physical and emotional needs. However, in real-world healthcare environments, communication is not always straightforward, especially when interacting with deaf or hard-of-hearing individuals.

Research has shown that deaf signers frequently experience reduced access to quality healthcare services due to communication barriers. For example, Jacob et al. highlight that many healthcare providers lack sufficient sign-language communication skills, which can lead to misunderstandings and poorer health outcomes [1]. Similarly, Abou-Abdallah and Lamyman emphasize that communication difficulties can negatively impact diagnosis accuracy, patient satisfaction and the effectiveness of care delivery [2]. This issue is not limited to a specific region as millions of individuals worldwide rely on sign language as their primary means of communication.

Despite this need, most caregivers are not formally trained in sign language, and even when training is available, it is often general-purpose and not tailored to caregiving contexts. Existing solutions attempt to address this gap, but they remain limited in several ways. For instance, platforms such as NVIDIA Signs provide AI - supported sign learning through webcam-based interaction and sign recognition. While this represents an important advancement in accessible learning, it primarily focuses on general vocabulary learning and data contribution, rather than guided, caregiver-specific tutoring [3]. Similarly, systems like Signing for Care introduce healthcare-related sign learning; however, they still report challenges such as limited personalization and unclear feedback [4].

These limitations highlight a key issue: current solutions tend to focus on recognition or content delivery, rather than effective learning and skill development. They often provide simple correct/incorrect feedback or static demonstrations, which are insufficient for building practical communication skills, especially for caregivers who require confidence and accuracy in real-time interactions [5], [6].

To address this gap, this research proposes a Personalized Sign Language Tutoring System for Caregivers. The system is designed to support caregivers in learning care-related signs through an adaptive and interactive approach. It combines computer vision-based gesture capture, speed-tolerant motion evaluation, interpretable corrective feedback, and 3D avatar-based visual guidance. Additionally, the system incorporates adaptive practice mechanisms that personalize learning based on individual performance and repeated errors. By focusing on learning rather than recognition, the proposed solution aims to improve both skill and retention in practical caregiving scenarios.

This research aligns SDG 3: Good Health and Well-being by improving the quality and inclusiveness of healthcare communication. It also contributes to SDG 10: Reduced Inequalities, as it addresses communication barriers faced by deaf and hard-of-hearing individuals. Furthermore, by leveraging intelligent technologies to support caregivers, the research indirectly supports SDG 4: Quality Education, through accessible and personalized learning.

This work lies at the intersection of multiple domains, forming a multidisciplinary research cluster. It integrates concepts from computer vision (gesture recognition and motion analysis), human-computer interaction, explainable AI and adaptive learning systems through personalized education. The combination of these domains moves beyond traditional sign recognition and contributes toward the development of intelligent, learner-centered tutoring systems for real-world applications.

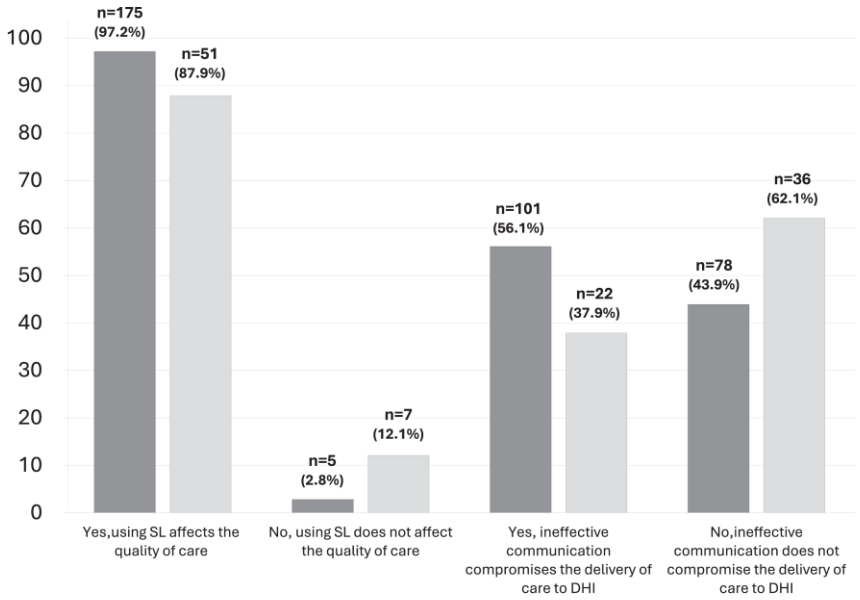


Figure 1 Effect of use of SL on quality of care and compromise Hearing Impaired patients care delivery from https://peerj.com/articles/19446/

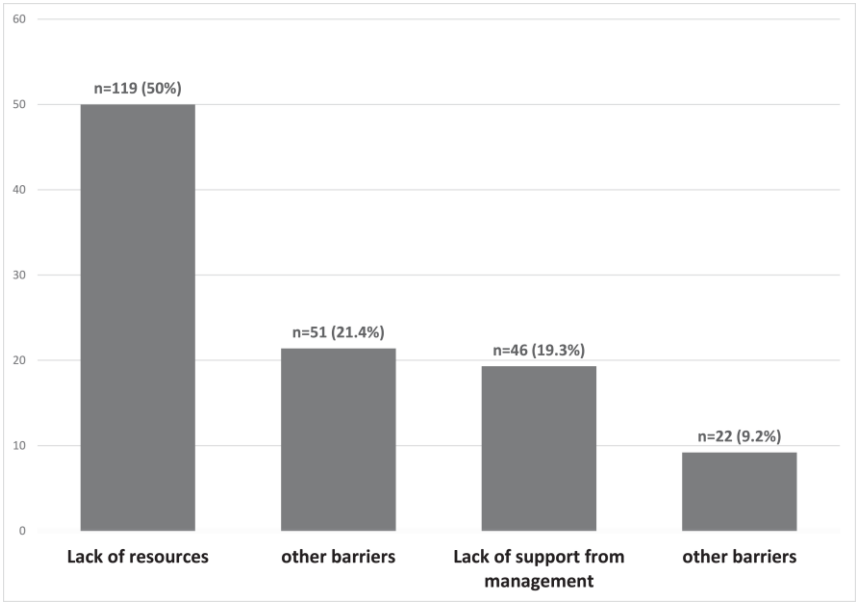


Figure 2 Perceived barriers to learning SL among hearing impaired patients from https://peerj.com/articles/19446/

1.2. Literature Review

Recent research shows that sign-language technology has gradually moved beyond simple sign recognition toward more interactive learning, feedback, and accessibility-oriented systems. This shift is especially important because communication barriers between care providers and deaf signers continue to affect the quality and trustworthiness of care delivery [1], [2]. Existing healthcare-focused studies consistently show that many providers are not sufficiently prepared to communicate through sign language, which highlights the need for systems that do more than display signs, and instead support practical skill development for caregivers [1], [2].

One major research approach in this area focuses on domain-specific sign-learning tools. For example, Signing for Care introduced an AI-supported ASL learning tool for emergency medical service providers, showing that healthcare-oriented sign instruction is both relevant and feasible [4]. However, the same work also revealed that domain-specific vocabulary alone is not enough, since learners still need clearer corrective feedback, improved pace control, and more personalized support [4]. This suggests that healthcare sign tutoring should not only teach vocabulary but also guide learners in how to perform signs correctly and confidently in practice.

A second research direction emphasizes learner feedback and sign-performance correction. Huenerfauth et al. demonstrated that learners benefit more when feedback is directly connected to their own signing attempts and supported by clear visual guidance rather than simple right-or-wrong judgment [5]. Similarly, Knapp and Bohacek showed that pose-aware AI interfaces can generate more interpretable written feedback from learner and reference sign videos, suggesting a promising shift toward explainable tutoring systems [6]. These studies indicate that effective sign-language learning systems should move beyond recognition and incorporate correction-oriented, learner-centered feedback. However, most existing approaches still remain either experimental or limited to isolated feedback generation rather than fully integrated tutoring environments [5], [6].

Another important literature focuses on the technical modeling of sign language as spatiotemporal motion. Approaches such as skeleton-aware multimodal recognition and monocular 3D hand pose estimation show that hand movement, body pose, and temporal dynamics can be captured effectively using computer vision methods [7], [8]. These methods provide a strong technical foundation for sign analysis, but their main objective is often recognition accuracy rather than educational support. In parallel, studies on avatar-based sign presentation and monitoring show that sign-language avatars can support learning, but only when motion quality, timing, non-manual cues, and user-centered design are carefully addressed [9], [10]. This means that while technical sign capture and avatar generation are increasingly advanced, they are not yet sufficient on their own to form an effective caregiver tutoring system.

In terms of existing products and services, platforms such as NVIDIA Signs represent an important advancement in AI-supported sign learning through webcam-based practice, real-time interaction, and avatar-based demonstration [3]. Likewise, interactive systems such as PopSignAI and collaborative learning approaches such as CoSignPlay demonstrate how

gamification and participation can make sign learning more engaging [11], [12]. However, these systems are largely designed for general sign-learning or engagement-focused contexts rather than caregiver-specific communication training.

Across the literature, a common limitation remains: most existing systems are either recognition-focused instead of tutoring-focused, general-purpose instead of caregiver-oriented, or interactive without sufficient personalization and adaptive reinforcement [3], [4], [9], [11]. Therefore, there remains a clear research gap for a personalized sign language tutoring system for caregivers that combines care-related sign learning, speed-tolerant evaluation, interpretable feedback, avatar-based visual guidance, and adaptive practice within a single integrated platform.

1.3. Research Gap

Although recent work has made important progress in sign-language technology, a key research gap remains: the absence of an integrated, caregiver-specific sign language tutoring system that supports adaptive learning and interpretable feedback. Existing studies show that communication barriers between healthcare providers and deaf signers continue to affect care quality, trust, and accessibility [1], [2]. At the same time, current sign-learning systems do not adequately address the specific learning needs of caregivers, who require practical, care-related sign communication rather than general vocabulary exposure.

Current approaches are insufficient for several reasons. First, healthcare-oriented systems such as Signing for Care demonstrate the value of domain-specific sign learning, but they still report limited personalization, unclear feedback, and difficulty distinguishing whether errors come from the learner or the system [4]. Second, feedback-centered studies show that learners benefit from guidance linked directly to their own signing attempts, especially when accompanied by visual correction [5], [6]. However, these approaches are mainly focused on feedback experiments or prototype interfaces, rather than complete tutoring systems that support continuous learning in real care contexts. Third, vision-based and multimodal sign recognition methods have improved technical performance in capturing hand and body motion [7], [8], yet their primary objective remains recognition accuracy, not learner support, error explanation, or adaptive tutoring.

A further limitation is that existing interactive and avatar-based systems still do not fully support the type of guided, personalized learning needed for caregivers. NVIDIA Signs provides webcam-based sign practice, real-time interaction, and avatar guidance, but it remains a general-purpose ASL learning platform rather than a caregiver-specific tutoring system [3]. Likewise, avatar-related studies and sign generation systems show that visually realistic sign presentation is becoming more advanced, but they also emphasize unresolved issues such as motion fidelity, non-manual cues, timing, and user-centered evaluation [9], [10]. Gamified and collaborative systems such as PopSignAI and CoSignPlay improve engagement [11], [12], but they are not designed to deliver care-related sign practice with interpretable corrective feedback and adaptive reinforcement.

Therefore, the key missing element in the current literature is a unified, caregiver-specific tutoring framework that goes beyond recognition and static content delivery. There is still no complete solution that combines care-related sign selection, real-time gesture capture, speed-tolerant sign evaluation, interpretable corrective feedback, 3D avatar-based visual guidance, and adaptive practice within a single web-based learning system. This gap is both practical and methodological: practical because caregivers need targeted communication support in real care settings, and methodological because current systems rarely separate sign correctness from signing speed, or translate gesture errors into learner-friendly corrections.

The novelty of this research lies in both its conceptual and technical contributions. Unlike existing systems that primarily treat sign language as a recognition or content delivery problem, this work reframes it as an interpretable, learner-centered tutoring process. The proposed system introduces a kinematic modeling approach that represents caregiver-performed signs as

spatiotemporal hand-joint motion sequences and compares them with expert references using speed-invariant evaluation, allowing movement accuracy to be assessed independently of execution pace. This enables fair evaluation for novice learners and accommodates irregular learning patterns common in caregiving contexts.

Furthermore, the system incorporates explicit error decomposition to generate interpretable corrective feedback, combined with real-time 3D avatar guidance that visually links user motion to correct sign execution. In addition, adaptive practice mechanisms and structured repetition support engagement and retention, addressing limitations in existing systems that rely on static demonstrations or simple correctness feedback. Together, this integration of kinematic evaluation, explainable feedback, avatar-based guidance, and adaptive reinforcement represents a novel and comprehensive tutoring architecture for caregiver-focused sign language learning.

Gap	Current Limitation in Research	Proposed Solution in This Study
Lack of caregiver-specific sign language training	Existing systems are general-purpose and do not focus on care-related communication needs	Develop a caregiver-oriented tutoring system focused on practical, real-world care interactions
Absence of integrated tutoring systems	Current approaches are fragmented (recognition, feedback, or avatars separately), not unified	Build a single integrated platform combining capture, evaluation, feedback, and guidance
Limited personalization and adaptive learning	Systems lack adaptive pacing, learner-specific feedback, and performance tracking	Introducing adaptive practice and personalized feedback based on user performance and errors
Lack of interpretable corrective feedback	Many systems provide simple correct/incorrect output without explaining errors	Implement interpretable feedback with clear error explanation and correction guidance
No separation of speed and accuracy in evaluation	Existing models evaluate signs rigidly without accounting for beginner speed variation	Use speed-tolerant (DTW-based) evaluation to assess motion independently of execution speed
Weak integration of visual learning support	Avatar systems lack tight linkage with user performance and feedback	Provide real-time 3D avatar guidance aligned with user motion for visual correction
Engagement without meaningful learning (gamification gap)	Gamified systems improve engagement but lack depth in feedback and skill development	Combine adaptive reinforcement + structured practice for both engagement and learning retention

Table 2Comparison of Research Limitations and Study Objectives

2. OBJECTIVE

2.1. Main Objective

The main objective of this research is to design and develop a personalized sign language tutoring system for caregivers that enables effective learning of care-related signs through adaptive, interpretable, and real-time feedback mechanisms. The system aims to support caregivers in acquiring practical communication skills by integrating computer vision-based gesture capture, speed-tolerant sign evaluation, and 3D avatar-guided correction within a unified web-based platform.

This objective is measurable through the system's ability to accurately evaluate user-performed signs, provide clear and actionable corrective feedback, and demonstrate improvement in user performance, usability, and learning outcomes during evaluation.

2.2 Specific Objectives

To identify and define a set of care-related sign language gestures and phrases required for routine caregiving communication

To develop a computer vision-based module that captures and evaluates caregiver-performed signs using hand landmark extraction techniques.

To implement a speed-tolerant gesture evaluation mechanism that distinguishes movement accuracy from execution speed.

To generate interpretable corrective feedback with real-time 3D avatar guidance to support effective learning and self-correction.

To design and integrate an adaptive learning mechanism that personalizes practice based on user performance and repeated errors.

To evaluate the system's usability, accuracy, and learning effectiveness through user-based testing and performance analysis.

3. METHODOLOGY

3.1. Research approach

This study will adopt a Design Science Research (DSR) approach combined with experimental evaluation. The Design Science approach will be used to design and develop a functional artifact, namely a personalized sign language tutoring system for caregivers, aimed at addressing real-world communication challenges in care delivery.

In addition, an experimental approach will be employed to evaluate the effectiveness of the proposed system in terms of usability, learning improvement, and accuracy of sign evaluation. This will ensure that the solution is not only technically feasible but also practically effective.

3.2. Technical methods and algorithms

The proposed system will follow a structured pipeline for gesture capture, evaluation, and feedback:

Hand Landmark Extraction: MediaPipe will be used to extract 2D/3D hand landmarks from real-time video input.

Kinematic Representation: The extracted landmarks will be transformed into spatiotemporal motion sequences, including joint angles, relative positions, and movement trajectories.

Speed-Tolerant Evaluation: Dynamic Time Warping (DTW) will be applied to compare user-performed gestures with expert reference sequences while accommodating variations in signing speed.

Error Detection: Differences between user and reference gestures will be analyzed to identify specific errors in handshape, movement, and timing.

Interpretable Feedback: Rule-based mechanisms will be used to generate clear, human-understandable corrective feedback based on detected errors.

3D Avatar Guidance: A 3D avatar will be integrated to demonstrate correct gestures and enable visual comparison with user performance.

Adaptive Learning Mechanism: The system will track user performance over time and adjust practice difficulty, repetition, and feedback based on recurring errors.

3.3. Tools and platforms with justification

Component	Technology / Tool	Proposed Usage
Programming Language	Python	Python will be used as the primary programming language to implement the system's backend, including gesture processing, landmark analysis, speed-tolerant evaluation, adaptive learning logic, and feedback generation. It is selected due to its strong support for computer vision and rapid prototyping.
Computer Vision & Gesture Analysis	MediaPipe	MediaPipe will be used to detect and track hand landmarks from real-time webcam input. It will enable efficient extraction of 21 hand-joint coordinates required for gesture analysis without requiring additional model training.
	OpenCV	OpenCV will be used for video capture, frame processing, and integration of webcam input into the gesture analysis pipeline.
	Dynamic Time Warping (DTW)	Dynamic Time Warping will be used to compare user-performed gesture sequences with expert reference gestures while allowing variation in execution speed, enabling fair evaluation for beginner users.
Feedback & Learning Logic	Rule-Based Feedback Engine	A rule-based approach will be implemented to interpret gesture differences and generate simple corrective feedback such as direction, position, or timing errors.
	Adaptive Learning Logic	A performance tracking mechanism will be implemented to adjust practice sessions dynamically based on repeated errors and user progress.
Frontend Development	HTML / CSS / JavaScript	These technologies will be used to design and structure the user interface of the system.
	React JS	React JS will be used to build a responsive and interactive web-based interface tailored for caregivers.
3D Avatar & Visual Guidance	Three.js / React Three Fiber	These technologies will be used to render a 3D avatar that demonstrates correct gestures and supports visual comparison for users.
	Blender (Optional)	Blender may be used to prepare simple avatar models and animations if required.
Backend Framework	Flask	Flask will be used as a lightweight backend framework to manage communication between the frontend, gesture evaluation module, and feedback system.
Database	SQLite	SQLite will be used to store user performance data, gesture records, and practice history, as it is lightweight and suitable for prototype-level systems.
Design Tools	Figma	Figma will be used to design user-friendly UI/UX interfaces tailored for caregivers.
	Draw.io / Canva	These tools will be used to create system diagrams and architecture visuals for documentation purposes.
Version Control	Git / GitHub	Git and GitHub will be used for code management, version tracking, and team collaboration throughout the development process.

Table 3Summary of Proposed Technology Stack and System Components

3.4. Data Collection and Ethical Considerations

The proposed system will utilize a curated dataset of care-related sign language gestures, which will be collected through recorded videos of predefined signs performed under real-world conditions. Gesture data will be captured using standard webcams or smartphone cameras, incorporating variations in speed, orientation, and environmental conditions to reflect practical caregiving scenarios. The recorded videos will be processed using MediaPipe to extract hand landmark sequences for analysis.

Ethical considerations will be strictly followed throughout the data collection process. Informed consent will be obtained from all participants prior to data collection. To ensure privacy, no personally identifiable information such as facial data will be stored, and only abstracted hand landmark data will be used for processing and evaluation. All collected data will be securely stored and used solely for research and educational purposes, in compliance with institutional ethical guidelines.

3.5. Validation Strategy and Performance Metrics

The proposed system will be evaluated using a combination of technical validation and user-based evaluation to ensure both functional accuracy and practical usability. Technical performance will be assessed by measuring the accuracy of gesture comparison, consistency of Dynamic Time Warping (DTW) alignment, and reliability of error detection.

In addition, user-based evaluation will be conducted to assess usability and learning effectiveness. Metrics such as System Usability Scale (SUS), user satisfaction, and ease of interaction will be considered. Learning improvement will be measured through performance comparisons before and after system usage, including improvements in sign accuracy, reduction in repeated errors, and increased confidence in sign execution. The system will also be compared against basic non-adaptive approaches to demonstrate its effectiveness as a personalized tutoring solution.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

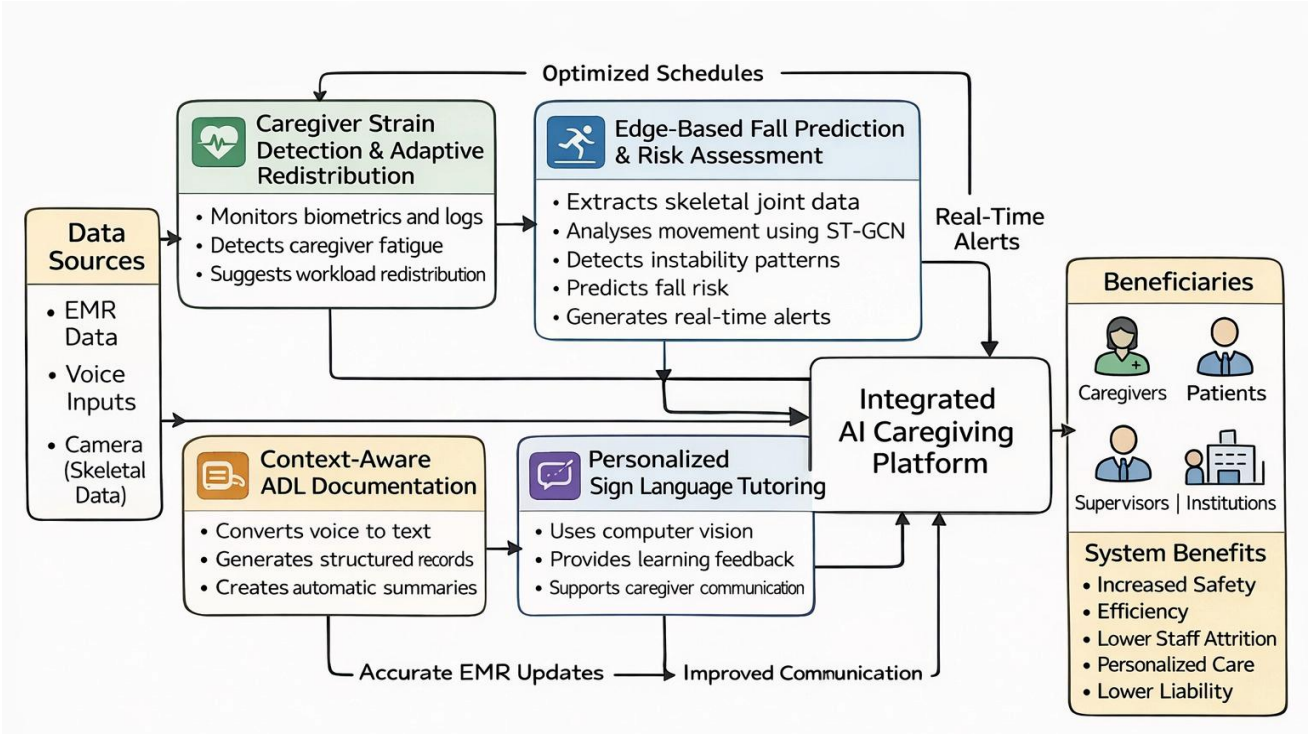


Figure 3 System Overview of the Integrated AI Caregiving Platform.

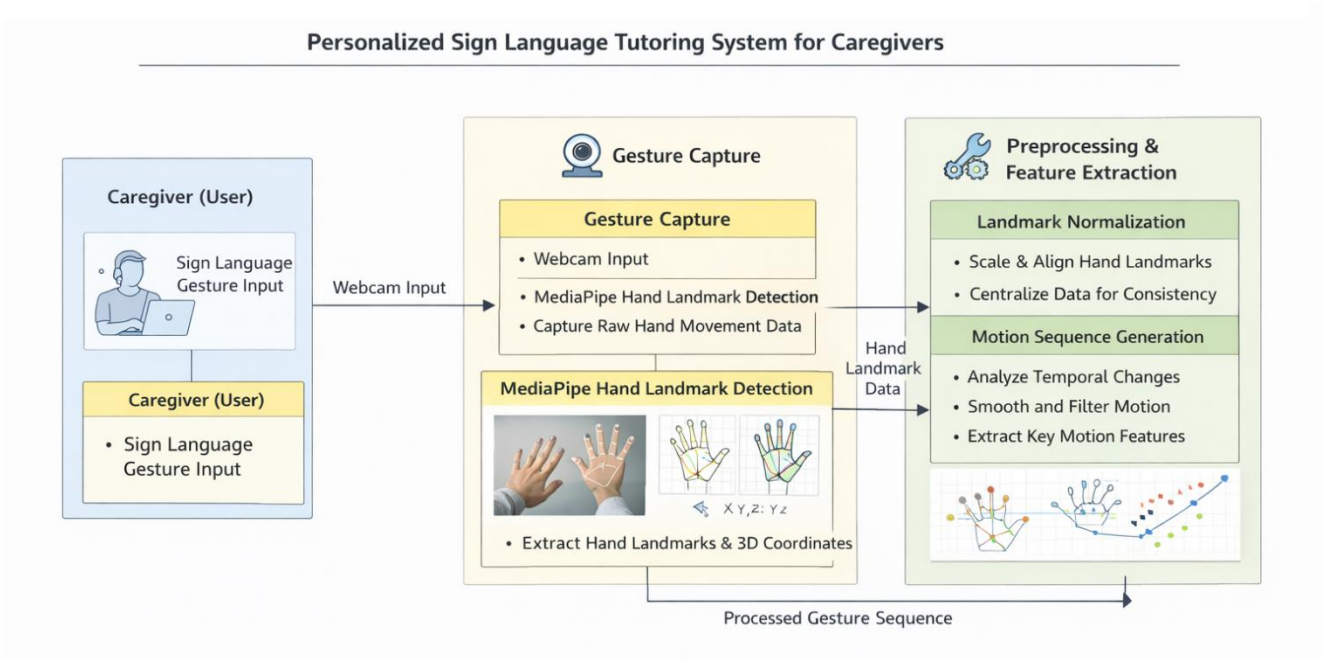


Figure 4 Component Specific Conceptual Diagram

The system diagram highlights the Personalized Sign Language Tutoring Component as a structured, modular pipeline consisting of key modules such as Gesture Capture, Preprocessing & Feature Extraction, Gesture Evaluation, Error Detection, Feedback Generation, 3D Avatar Guidance, and Adaptive Learning. Each module plays a specific role in transforming raw webcam input into meaningful learning outcomes, enabling accurate gesture analysis, interpretable feedback, and personalized skill development for caregivers.

The diagram also illustrates the data flow and component interactions, where gesture data captured from the caregiver is processed into hand landmark sequences, normalized, and converted into spatiotemporal motion features. These are then compared with reference sign data using a speed-tolerant evaluation mechanism to identify errors related to handshape, movement, and timing. The results are passed through the feedback module to generate actionable corrections, which are visually supported through 3D avatar guidance and reinforced through adaptive learning based on user performance and repeated errors.

Additionally, the diagram incorporates scalability and security considerations to ensure practical deployment. The modular design allows the system to scale by adding new signs, users, and learning features without affecting core functionality. The web-based architecture supports accessibility across devices, while privacy is maintained by processing only abstracted hand landmark data instead of storing personal information. Secure handling of user performance data and the ability to integrate with external systems ensure that the solution is scalable, secure, and suitable for real-world caregiving environments.

5. USER REQUIREMENTS

5.1. Stakeholder identification

The primary stakeholders of the proposed system include caregivers, supervisors or ward managers who are the main users and will utilize the system to learn and practice care-related sign language for effective communication with deaf or hard-of-hearing individuals. Deaf care recipients are indirect stakeholders, as they benefit from improved communication and quality of care.

Secondary stakeholders include healthcare institutions and training providers, who may adopt the system as a learning tool for caregiver training, and researchers/developers, who may use the system for further improvements in sign language tutoring and assistive technologies.

5.2. Functional requirements

The system will provide the following core functionalities:

- The system shall allow users to access and select care-related sign language lessons and gestures.
- The system shall enable users to capture hand gestures in real-time using a webcam.
- The system shall extract and process hand landmarks from captured gestures using computer vision techniques.
- The system shall compare user-performed gestures with predefined reference gestures using a speed-tolerant evaluation method.
- The system shall identify errors in gesture execution, including handshape, movement, and timing differences.
- The system shall provide interpretable corrective feedback to guide users on how to improve their signing.
- The system shall display 3D avatar-based demonstrations of correct gestures for visual guidance.
- The system shall support adaptive learning by tracking user performance and adjusting practice accordingly.
- The system shall store user performance data and practice history for progress monitoring.
- The system shall provide a user-friendly interface for easy navigation and interaction.

5.3. Non-functional requirements

The proposed system will be designed to ensure high levels of usability, performance, accuracy, and reliability, while also maintaining security and accessibility. From a usability perspective, the system will provide an intuitive and user-friendly interface that can be easily used by caregivers with minimal technical knowledge. The interface will be designed to support simple navigation, clear instructions, and interactive feedback to enhance the overall learning experience.

In terms of performance, the system will support real-time gesture capture and feedback with minimal latency, ensuring smooth interaction during sign practice sessions. Accuracy will be a key requirement, as the system will provide reliable gesture evaluation and precise error

detection to support effective learning and skill improvement. Additionally, the system will be designed with scalability in mind, allowing future expansion to include more sign language content, additional users, and enhanced features without major structural changes.

Security and privacy will be carefully considered by ensuring that no personally identifiable information is stored and that all user data is handled securely. The system will only process abstracted hand landmark data to maintain user anonymity. Furthermore, the system will be accessible through standard web browsers, enabling use across multiple devices without requiring specialized hardware. Finally, the system will ensure reliability by functioning consistently during repeated use, minimizing errors, system crashes, or interruptions during operation.

6. COMMERCIALIZATION PLAN

6.1. Target market and customer persona

The primary target market includes healthcare institutions, caregiver training centers, and home-care service providers. The system is designed for caregivers with little or no prior sign language knowledge who require practical communication skills in daily care settings. The typical user persona includes busy caregivers seeking quick, accessible, and practical learning tools that can be integrated into their workflow without extensive training.

6.2. Value proposition

The proposed system offers a caregiver-specific, personalized sign language tutoring solution that goes beyond general sign-learning tools. It provides real-time gesture evaluation, interpretable feedback, adaptive learning, and 3D visual guidance, enabling caregivers to learn practical signs efficiently and confidently. This directly improves communication quality, inclusiveness, and care delivery outcomes.

6.3. Revenue model

The system can adopt a hybrid revenue model combining institutional deployment and scalable service offerings. A one-time deployment model can be offered to healthcare institutions and caregiver training centers, where the system is installed, customized, and integrated into their internal training programs.

In addition, a Software-as-a-Service (SaaS) model can be implemented, providing subscription-based access with continuous updates, technical support, and cloud-based features such as performance tracking, analytics, and remote learning capabilities.

Furthermore, the system can introduce a training and certification model, where caregivers can complete structured learning modules and receive certification for sign-language competence, generating additional revenue through course-based access.

An optional freemium model can also be considered, where basic sign-learning features are provided for free, while advanced features such as personalized feedback, adaptive learning, and detailed performance analytics are available through paid upgrades.

6.4. Cost estimation

The primary costs will include system development, testing, and maintenance, as well as infrastructure costs such as hosting and storage. Since the system is built using open-source technologies, software costs will be minimal. Additional costs may include UI design, dataset preparation, and deployment setup, making the overall solution cost-effective for both development and scaling.

6.5. Pricing strategy

The pricing strategy will be designed to ensure affordability and scalability.

- For healthcare organizations, a fixed one-time licensing fee will be offered for deployment.
- For smaller care clinics or centers, a tiered subscription model (basic, standard, premium) can be implemented based on features such as advanced analytics, personalization, and support.This flexible pricing approach ensures accessibility for both small-scale users and larger organizations.

6.6. Competitive advantage

The system's key competitive advantage lies in its integration of personalized tutoring, speed-tolerant evaluation, interpretable feedback, and caregiver-specific learning content. Unlike existing systems that focus on general sign recognition or vocabulary learning, this solution is task-oriented, adaptive, and learner-centered, making it more effective for real-world caregiving scenarios.

6.7. Intellectual property considerations

The proposed system may be protected through software copyrights, covering the overall implementation, system architecture, user interface design, and integration of the personalized tutoring platform. In addition, key innovative components such as the speed-tolerant gesture evaluation mechanism, kinematic motion modeling approach, interpretable feedback generation framework, and adaptive learning logic may be considered for further intellectual property protection, including potential patentability depending on novelty and feasibility.

The dataset of care-related sign gestures, including curated reference gesture sequences and structured learning content, may also be treated as proprietary assets, especially if manually developed or annotated for this system. Proper ownership and usage rights will be maintained for any internally created data.

All third-party libraries and frameworks, such as MediaPipe, OpenCV, and Three.js, will be used in compliance with their respective open-source licenses, ensuring that licensing terms are respected and no violations occur. The system will also ensure that no copyrighted sign language content is used without appropriate permission.

Furthermore, any user-generated performance data will be handled in accordance with data protection and privacy regulations, ensuring that ownership remains with the user and is not exploited for commercial purposes without consent. Future commercialization may also consider trademark protection for the system name and branding to establish a distinct identity in the market.

7. BUDGET AND JUSTIFICATION

Category	Item	Estimated Cost Range (LKR)	Description
Cloud Hosting & Deployment	Hosting Upgrade (Optional)	5,000 – 10,000	Short-term paid hosting for stable demonstration and testing.
Data Collection & Testing	Participant Compensation	8,000 – 15,000	Incentives for participants involved in usability testing and data collection.
	Recording Setup	3,000 – 7,000	Basic accessories such as lighting and stand for consistent data capture.
Development Effort	System Development (4 months)	20,000 – 30,000	Estimated development effort for implementing gesture evaluation, feedback system, UI, and integration.

Table 4Component Specific Budget and Estimation

Estimated Costs for Research Project

Description	Hours / Units	Cost (LKR)
Labour Costs		
Software Development	350 hrs	42,000
System Integration	250 hrs	30,000
Testing & Deployment	200 hrs	24,000
Total Labour Cost		**96,000**
Material & Infrastructure Costs		
USB Camera for Fall Detection	1 unit	12,000
Cloud Hosting & Storage (6 months)	–	10,000
Software Tools / APIs / GPU Usage (4 months)	–	8,000
Total Material & Infrastructure Cost		**30,000**
GRAND TOTAL		**126,000**

Figure 6 Budget Breakdown for the Research Project.

8. WORK BREAKDOWN STRUCTURE (WBS)

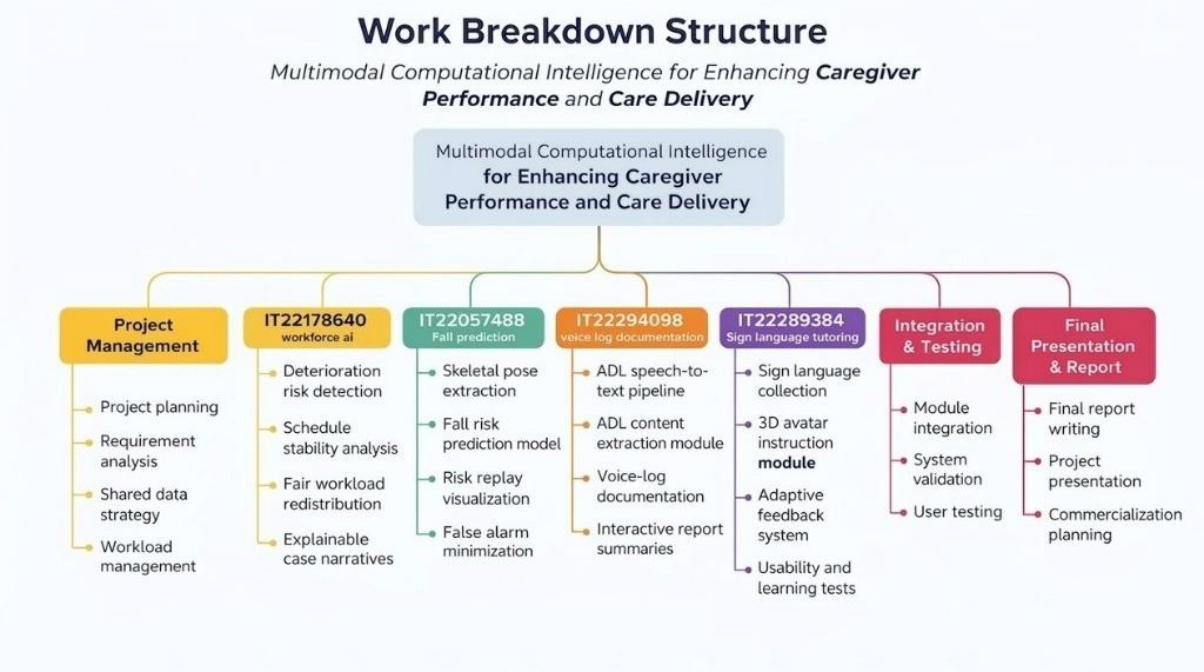


Figure 7 Overall Work Breakdown Structure

9. GANTT CHART

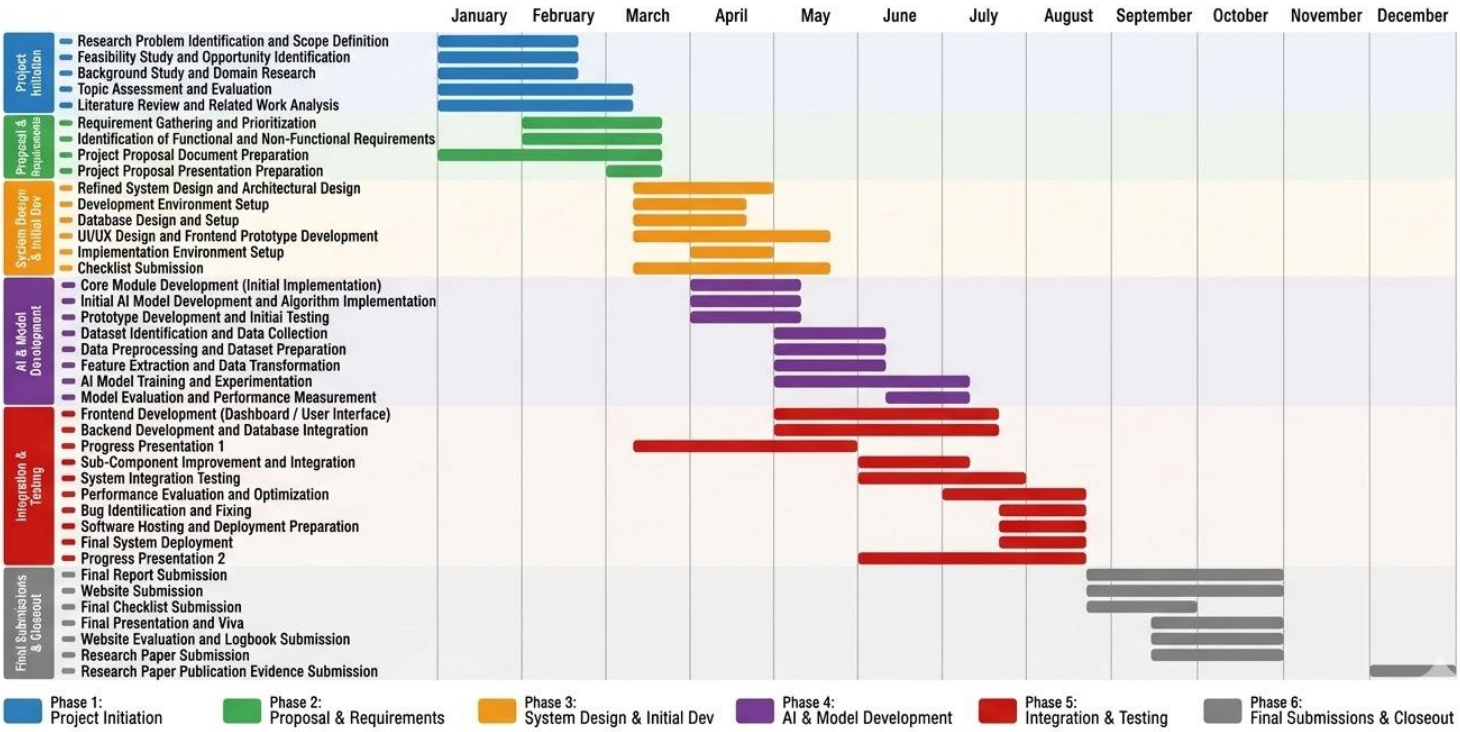


Figure 8 Project Timeline

10.AI DISCLOSURE

During the preparation of this document, AI tools were used in a supportive and limited manner, strictly in line with the guidelines of the Research Project AI Usage Policy. ChatGPT and Gemini AI tools were used primarily to enhance the clarity of writing, restructure sentences, and refine the presentation of ideas.

The development of all key components of this work including the research problem, proposed approach, design decisions, and technical direction was carried out independently. AI tools were not used to generate complete sections or to substitute original thinking, but rather to enhance the quality and readability of content that was already developed by the author.Any suggestions or outputs provided by AI were not accepted at face value. They were critically evaluated, adjusted where necessary, and verified against reliable academic and technical sources. Additional care was taken to ensure that the content remains accurate, relevant, and aligned with the intended research objectives.

Overall, AI usage remained minimal and supplementary. The author retains full ownership of the work and accepts complete responsibility for its accuracy, integrity, and originality.

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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Project Title: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery - Personalized Sign Language Tutoring System for Caregivers

Group ID: R26-DS-010

Signature

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• Date: 23/3/26

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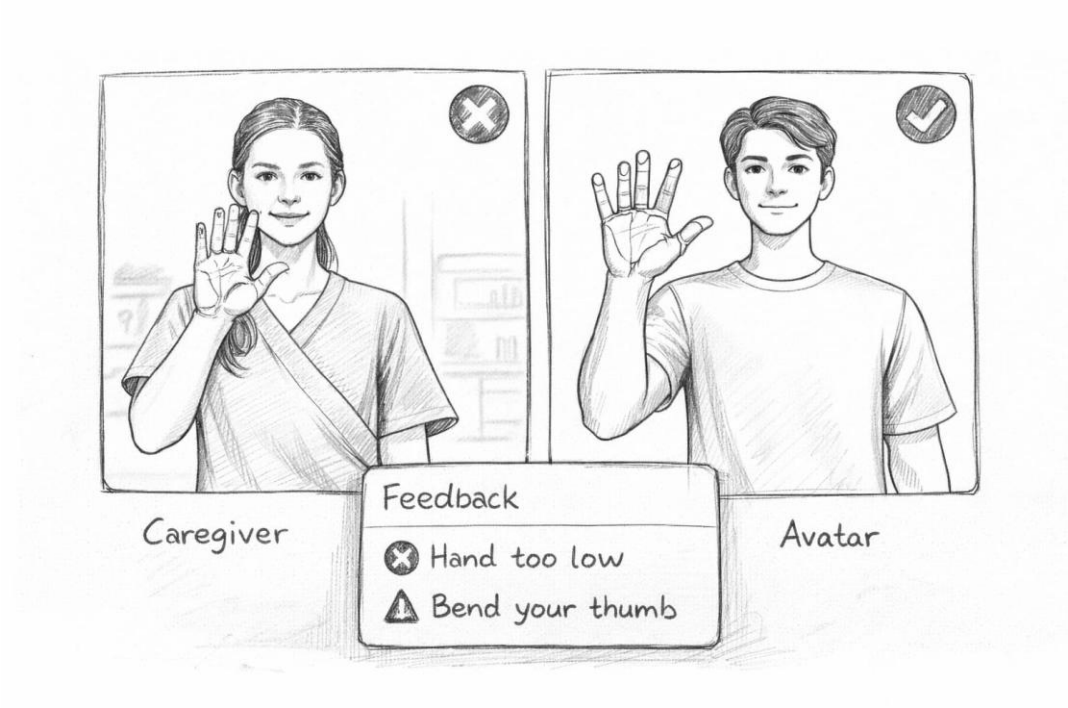
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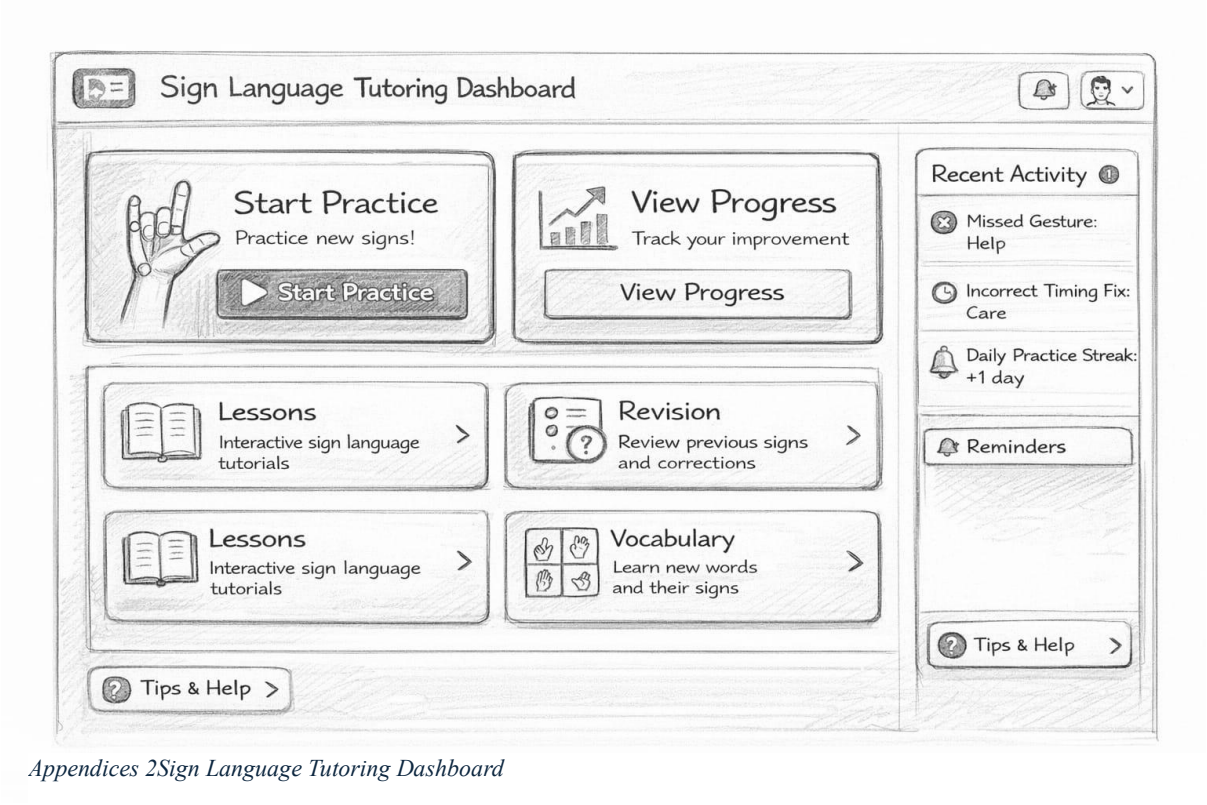
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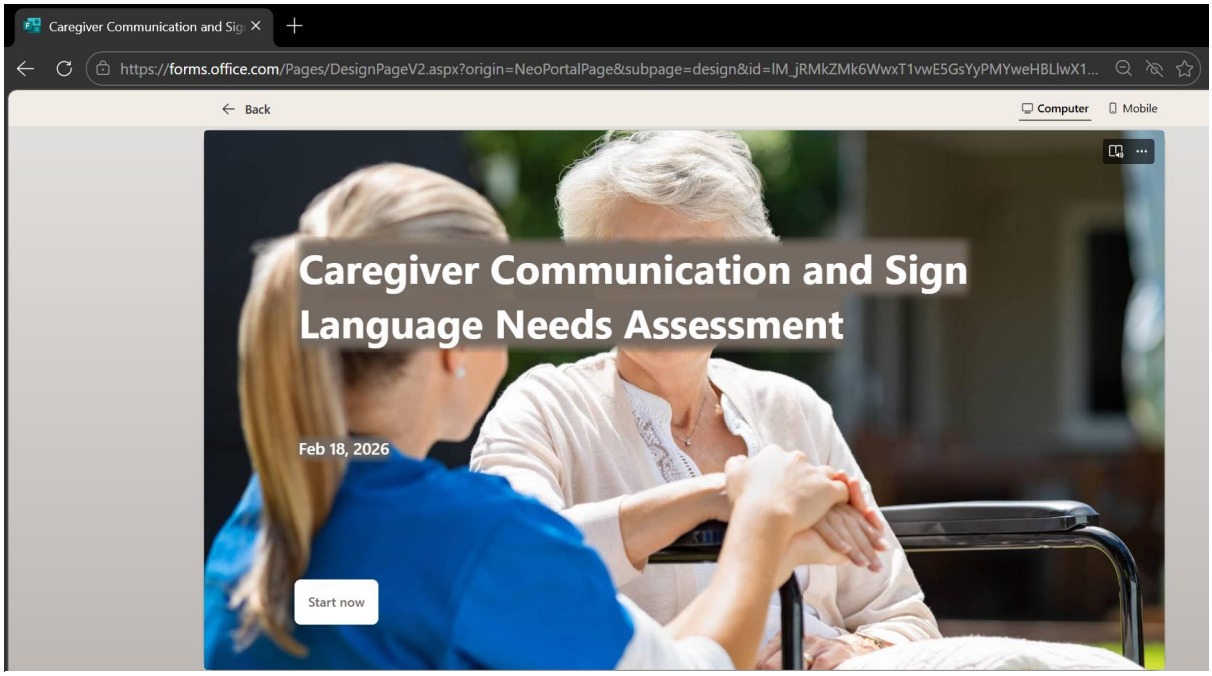
12. APPENDICES



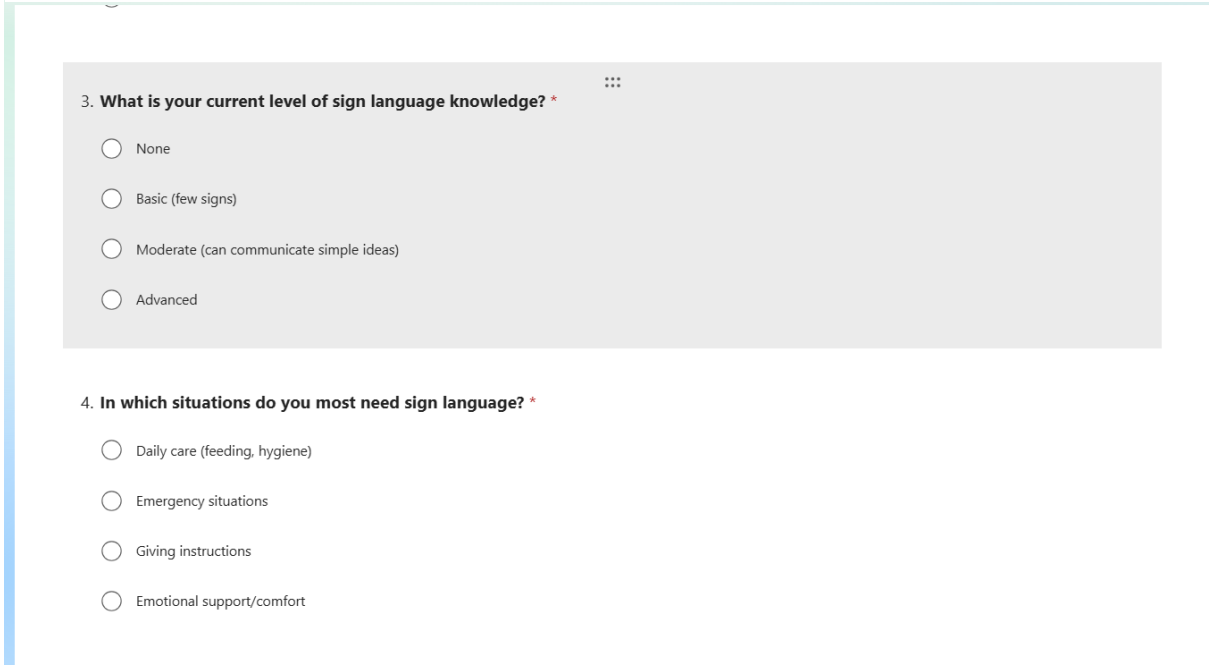
Appendices 1Caregiver practicing Sign Language with Real Time Feedback



Appendices 2Sign Language Tutoring Dashboard



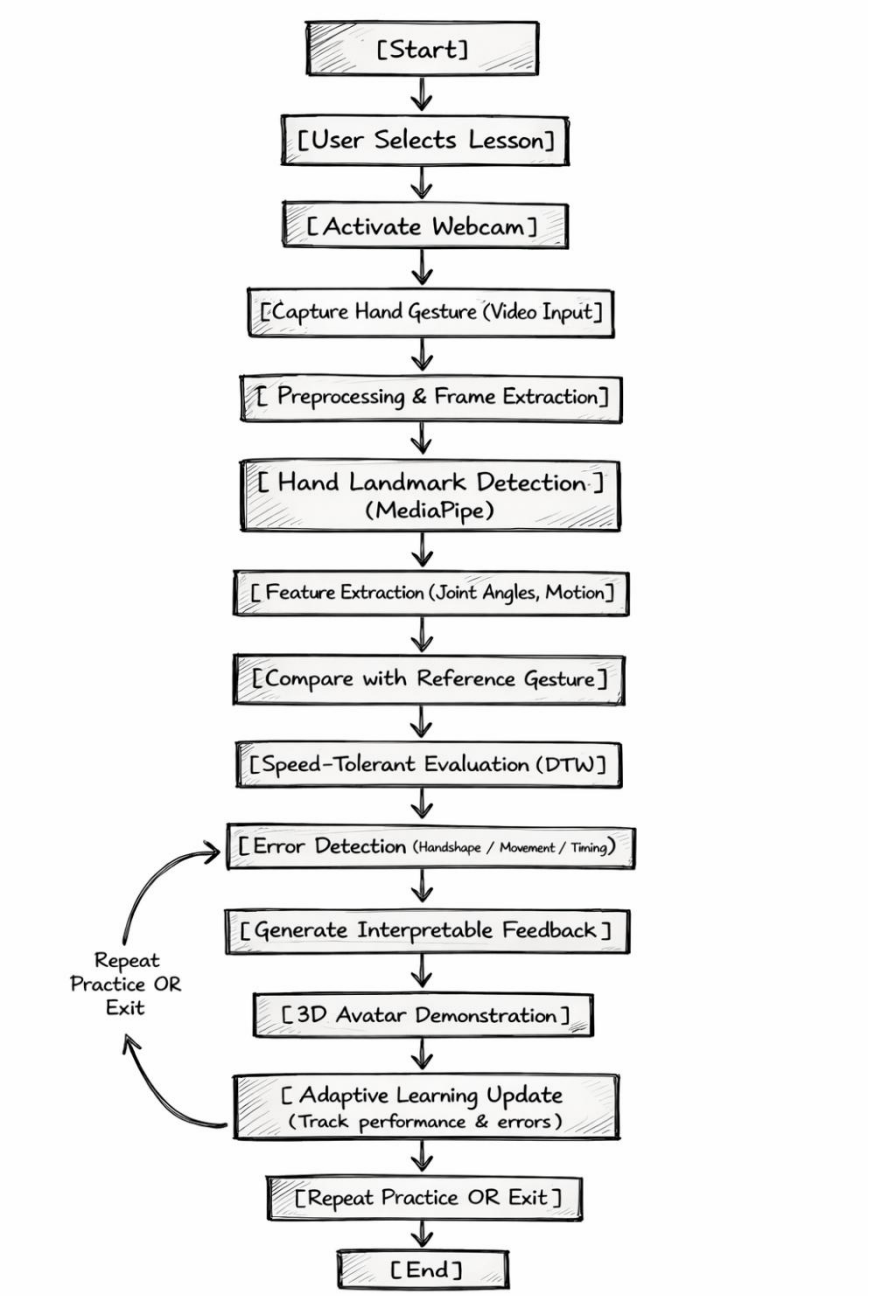
Appendices 3 Survey Landing Page



Appendices 4 Sample Survey Questions for Caregiver



Appendices 5 Survey Results Visualization



Appendices 6 Conceptual Workflow of Personalized Sign Language Tutoring System